Security Operations Center (SOC) Deployment Technical Report

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Project Title: ***Deployment and Validation of a Cloud-Based SOC Using Elastic SIEM***

Executive Summary

This project documents the deployment of a cloud-based Security Operations Center (SOC) leveraging Elastic SIEM on Elastic Cloud, integrated with AWS cloud telemetry. The SOC ingests multiple log sources including CloudTrail, VPC Flow Logs, and EC2 instance logs to provide comprehensive visibility into identity, network, and host activities. Detection rules, both out-of-the-box (OOTB) and a custom SSH activity rule, were implemented to monitor suspicious behavior. To validate the SOC, simulated events such as IAM user and group creation, SSH access, and ping tests were performed. Endpoint forensics was conducted on a Windows EC2 instance, and incidents were tracked, triaged, and documented. The result is a functional SOC capable of detecting and alerting relevant security events across AWS infrastructure.

Environment Setup

1.1 AWS Infrastructure:

* EC2 instances deployed in eu-north-1

* VPC with flow logging enabled

* S3 bucket configured as destination for VPC Flow Logs

* Windows instance deployed for endpoint forensics (RDP access)

1.2 Elastic Cloud Deployment:

* SIEM instance deployed in eu-west-2

* Elastic Agents installed on EC2 instances to ship logs

* OTX integration enabled for threat intelligence feeds

1.3 Log Sources Ingested:

* CloudTrail Logs

* VPC Flow Logs

* EC2 Instance Logs

* Windows Event Logs

Detection Engineering

2.1 Out-of-the-Box (OOTB) Rules

- AWS IAM Group Creation Rule Enabled
- AWS IAM User Addition to Group Rule Enabled
- Successful SSH Authentication from Unusal SSH Public Key Rule Enabled

2.2 Custom Rule

- Rule Name: Custom SSH Port 22 Connection
- Description: Detects SSH connections from new source IPs or abnormal activity patterns on EC2 instances

- Trigger Condition: Any SSH attempt outside predefined IP whitelist

- Outcome: Generates Medium-priority alert in Elastic SIEM

- **Additional Rules Enabled:** Suspicious IAM Role Usage, EC2 talking to multiple destinations, Successful SSH Authentication from Unusual User, Successful SSH Authentication from Unusual IP Address, AWS EC2 Instance Connect SSH Public Key Uploaded

Attack Simulation

- IAM Simulation: Created a new IAM user and group; CloudTrail alerts triggered **-**

SSH Simulation: Connected to EC2 via SSH from a new IP and conducted ping tests

- Timeline of Events:

Date & Time, Action, Detection/Alert

10 Feb 2026 01:52:46:000, Created and added IAM user ***'Shallipoppi'*** to Group, CloudTrail OOTB alert triggered

10 Feb 2026 01:52:14.000, Created IAM group ***'Casamigos'*** & attached ***AdministratorAccess*** policy, CloudTrail OOTB alert triggered

08 Feb 2026 17:19:11.000, Ping tests, Flow Logs captured and visualized

08 Feb 2026 17:16:01.624, SSH connection from new IP to EC2, Custom SSH detection triggered

Endpoint Forensics

- Deployed Windows EC2 instance

- Connected via RDP

- Pulled 'Log cleared' event (Event ID 1102)

Incident Report

Timeline:

- 10 Feb 2026 01:52:46:000 -> IAM user created and added to group

- 10 Feb 2026 01:52:14.000 -> IAM group created & policy attached

- 08 Feb 2026 17:35:42 -> Endpoint forensics collected - 08 Feb 2026 17:19:11.000 -> Ping tests

- 08 Feb 2026 17:16:01.624 -> SSH connection from new IP

Detection Methods:

- CloudTrail OOTB rules detected IAM modifications

- Custom SSH rule detected unexpected SSH access -

VPC Flow Logs visualized network traffic patterns -

Endpoint logs confirmed host-level events

Triage / Severity Assessment:

- IAM user creation outside normal business process -> High Severity

- SSH connection from unrecognized IP -> Medium Severity - Ping test -> Low Severity

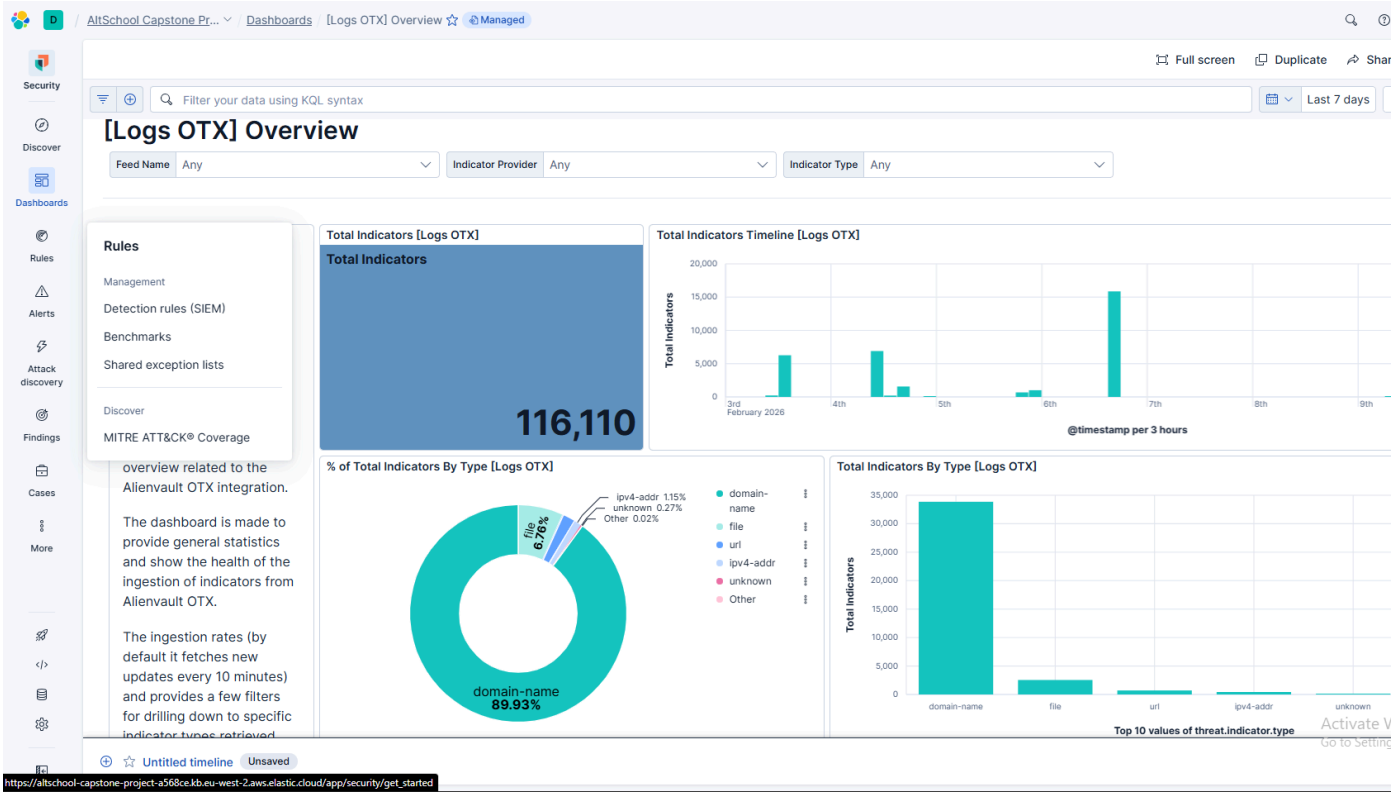
Response / Remediation:

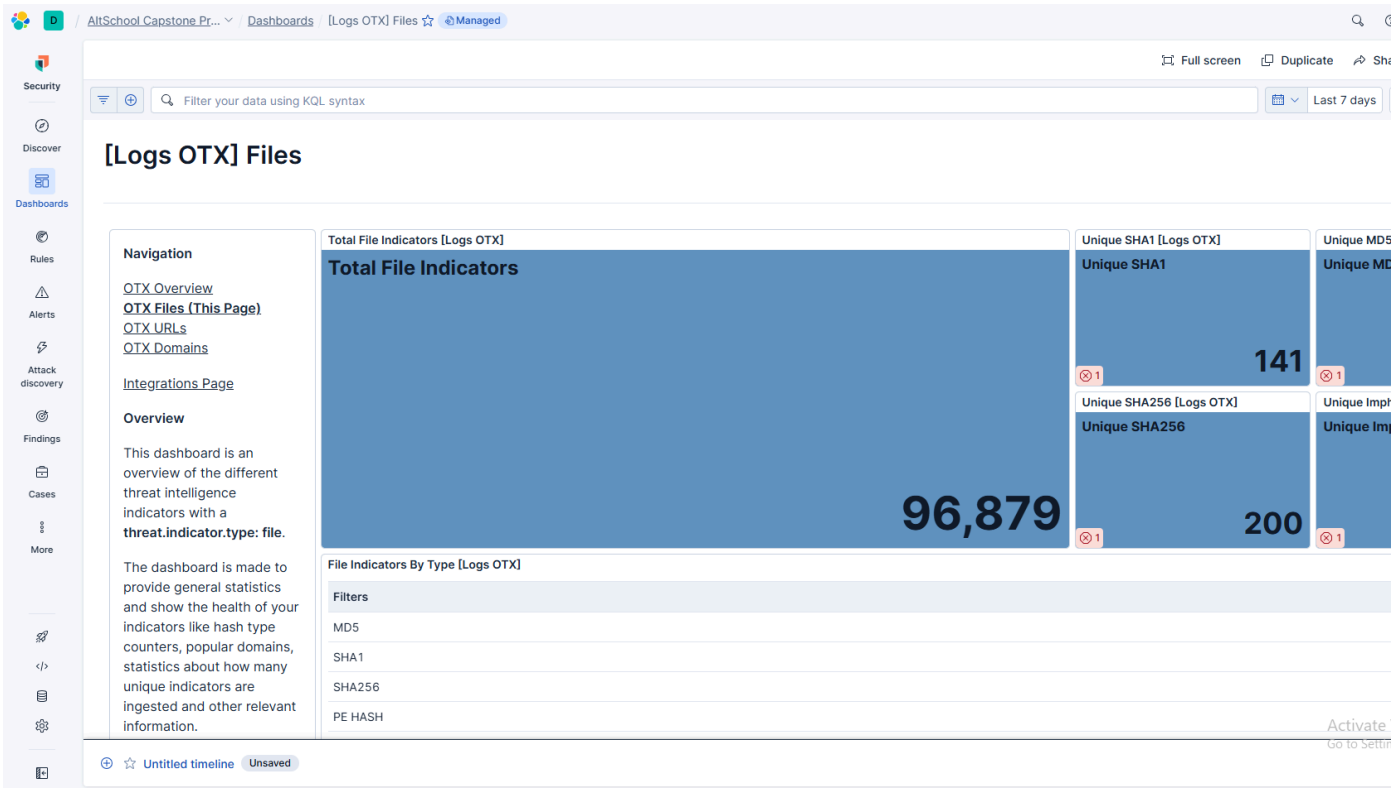
- Confirmed IAM activity, verified authorized user

- Blocked unrecognized SSH IP if unauthorized

- Ensured Flow Logs and Windows logs retained for audit

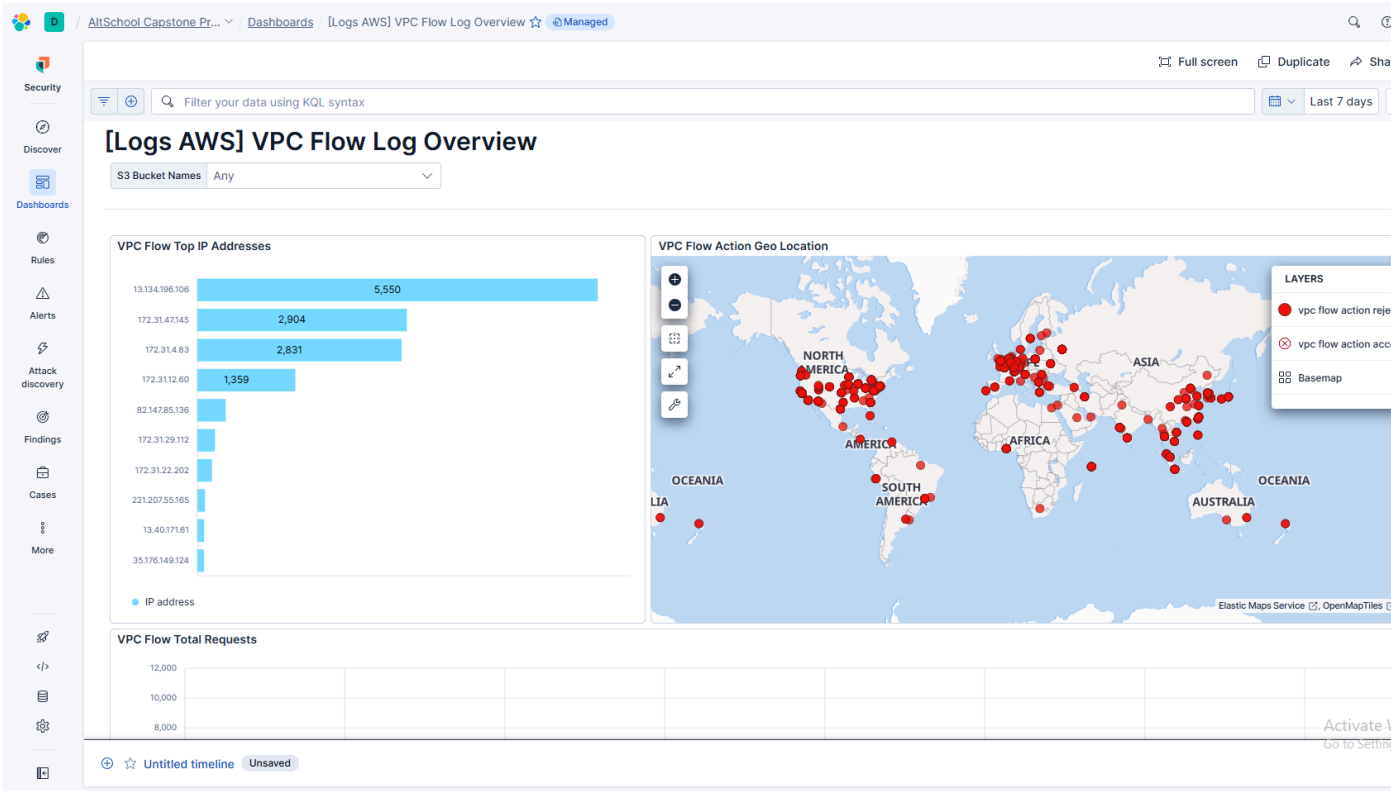
Appendix A

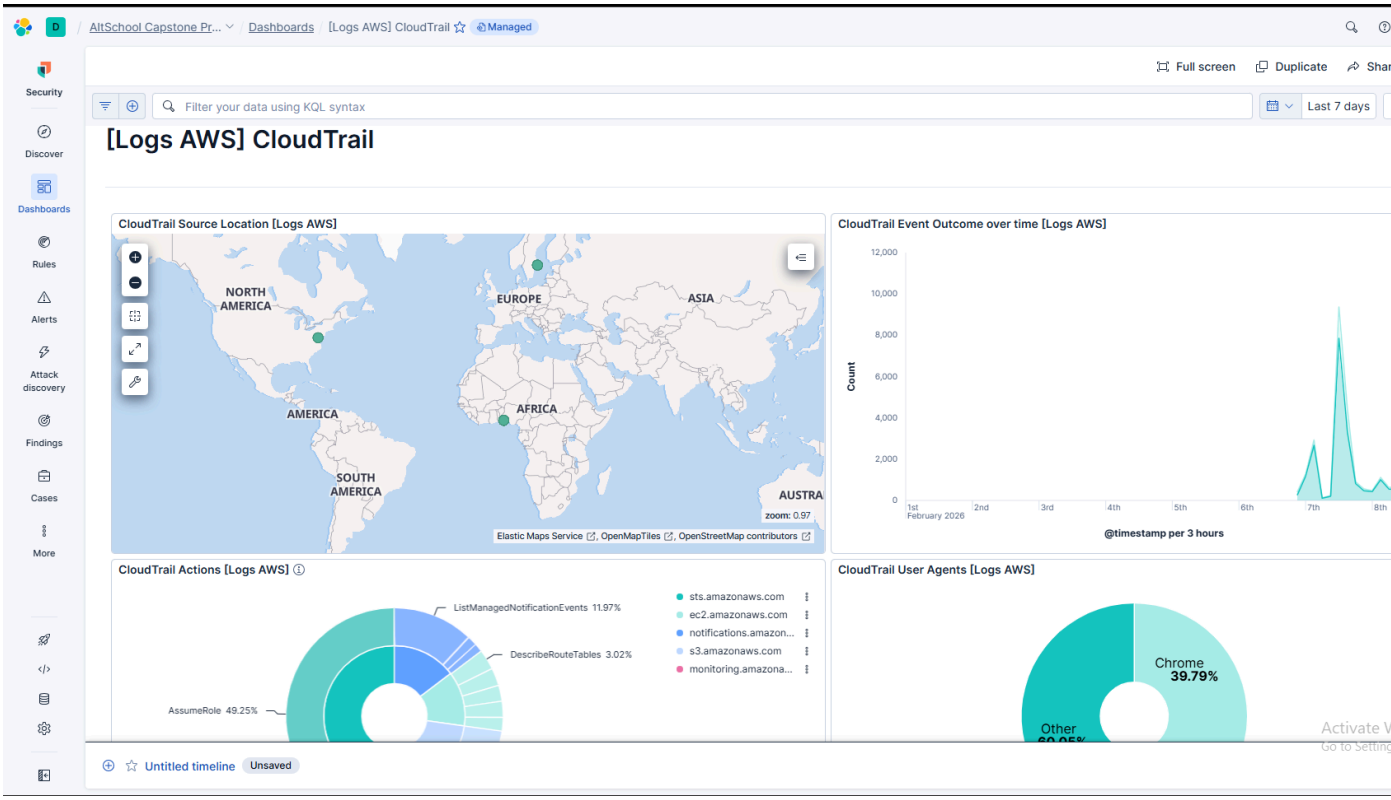




Evidence of successful AlienVault OTX Integration

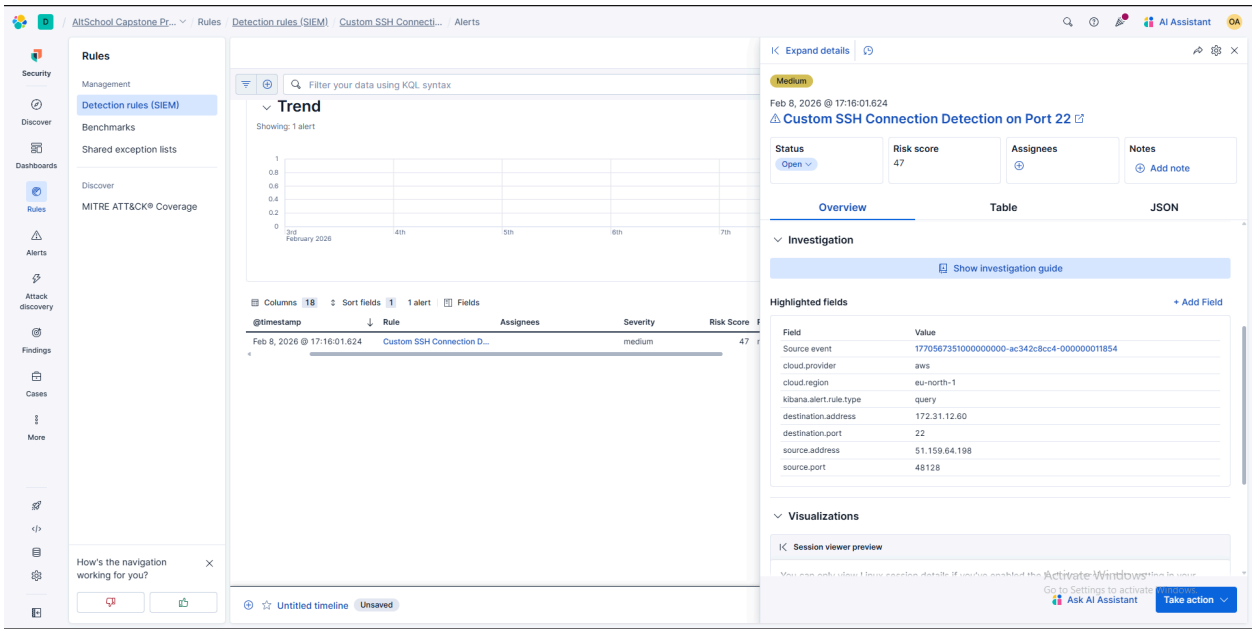
Appendix B





Dashboard views showing active Flow Logs and Audit Logs.

Appendix C


AltSchool Capstone Pr... Rules Detection rules (SIEM) Custom SSH Connecti... Alerts
AI Assistant
Security
Discover
Dashboards
Rules
Alerts
Attack discovery
Findings
Cases
More
Rules
Management
Detection rules (SIEM)
Benchmarks
Shared exception lists
Discover
MITRE ATT&CK® Coverage
How's the navigation working for you?
Filter your data using KQL syntax
Trend
Showing: 1 alert
Columns 18 Sort fields 1 1 alert Fields
@timestamp Rule Assignees Severity Risk Score
Feb 8, 2026 @ 17:16:01.624 Custom SSH Connection D... medium 47
Untitled timeline Unsaved
Expand details
Medium
Feb 8, 2026 @ 17:16:01.624
Custom SSH Connection Detection on Port 22
Status Open
Risk score 47
Assignees
Notes Add note
Overview Table JSON
Investigation
Show investigation guide
Highlighted fields + Add Field
Field Value
Source event 1770567351000000000-ac342c8cc4-000000011854
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cloud.region eu-north-1
kibana.alert.rule.type query
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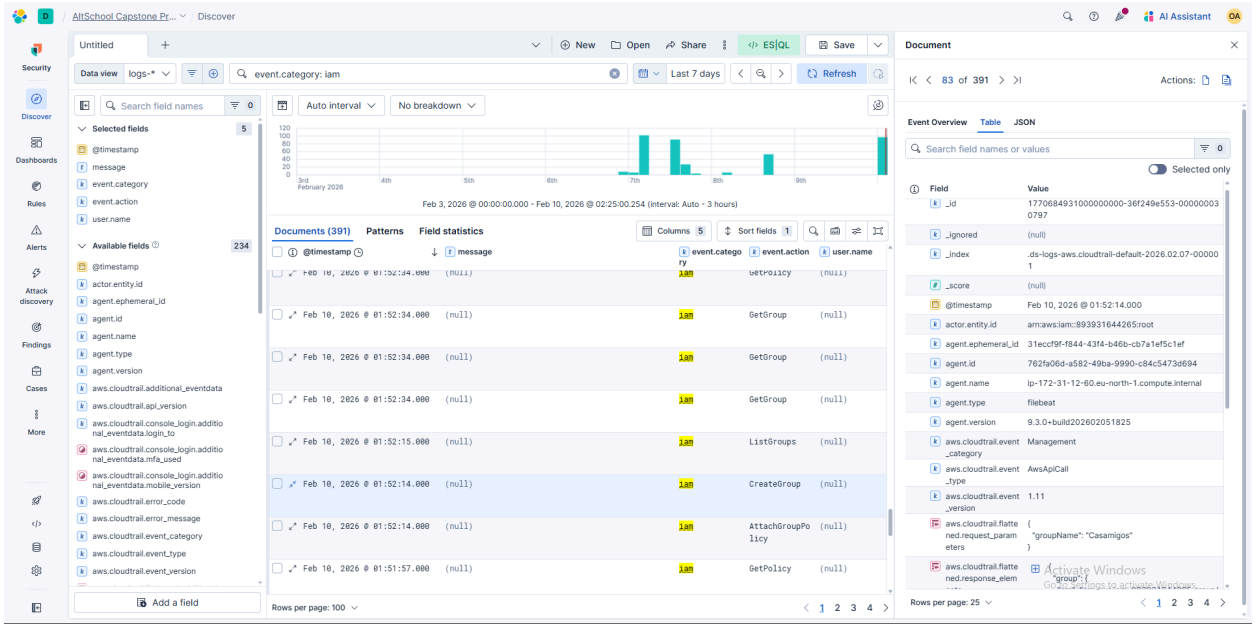
Visualizations
Session viewer preview
Ask AI Assistant
Take action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AI Assistant
Untitled
New Open Share ES|QL Save
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Search field names
Selected fields 5
@timestamp
message
event.category
event.action
user.name
Available fields 234
Auto interval No breakdown
Feb 3, 2026 @ 00:00:00.000 - Feb 10, 2026 @ 02:25:00.254 (interval: Auto - 3 hours)
Documents (391) Patterns Field statistics
Columns 5 Sort fields 1
@timestamp message event.category event.action user.name
Feb 10, 2026 @ 01:52:34.000 (null) iam GetGroup (null)
Feb 10, 2026 @ 01:52:15.000 (null) iam ListGroups (null)
Feb 10, 2026 @ 01:52:14.000 (null) iam CreateGroup (null)
Feb 10, 2026 @ 01:52:14.000 (null) iam AttachGroupPolicy (null)
Feb 10, 2026 @ 01:51:57.000 (null) iam GetPolicy (null)
Add a field
Rows per page: 100
Document
83 of 391
Actions:
Event Overview Table JSON
Search field names or values
Selected only
Field Value
_id 1770684931000000000-36f249e553-00000003 0797
_ignored (null)
_index .ds-logs-aws.cloudtrail-default-2026.02.07-00000 1
_score (null)
@timestamp Feb 10, 2026 @ 01:52:14.000
actor.entity.id arn:aws:iam::893931644265:root
agent.ephemeral_id 31eccf9f-f844-43f4-b46b-cb7a1ef5c1ef
agent.id 762fa06d-a582-49ba-9990-c84c5473d694
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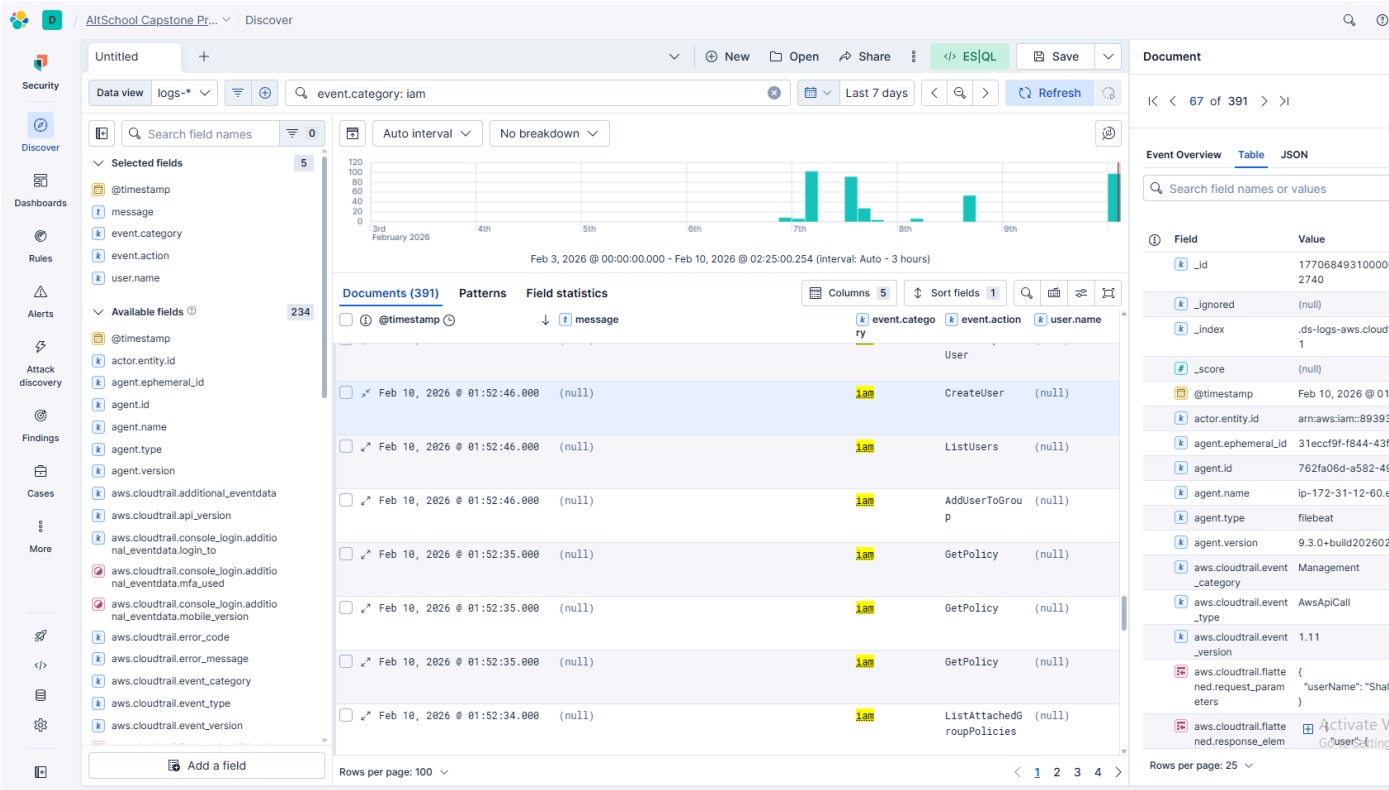
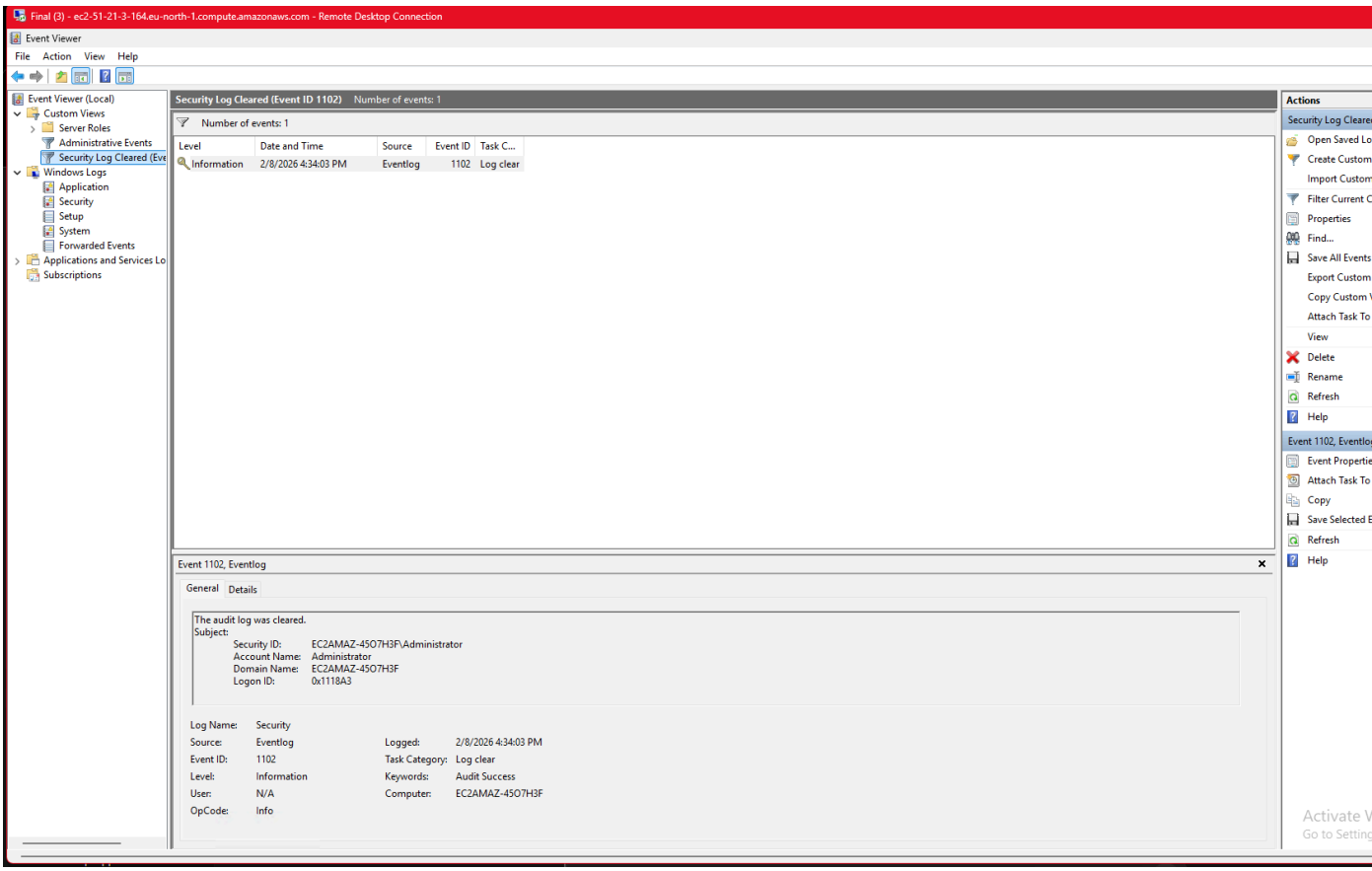


Evidence of triggered Out-of-the-Box and Custom SSH Rules

Appendix D



Windows Event Viewer screenshot showing Event ID 1102 (Log Cleared)

SUMMARY & CONCLUSION

In under 48 hours, a functional SOC environment was deployed integrating Elastic SIEM with AWS cloud telemetry. Logs from CloudTrail, VPC Flow Logs, and EC2 instances were ingested successfully, and detection rules-both OOTB and custom-applied. Attack simulations and endpoint forensics validated the SOC's ability to detect and alert on security-relevant events. This demonstrates proficiency in SOC design, detection engineering, incident analysis, and cloud-based threat monitoring.